COURSE PRACTICE MANUAL

ENTERPRISE RESOURCE PLANNING - ACCT5123

TOPIC

INTRODUCTION TO PRACTICAL CONTENT

Teaching assistant

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AGENDA

CONTENTS

1. General requirements
2. Introduction to the content of the practice sessions
3. Duration and grading ratio
4. Requirements for projects
5. Report content
6. General structure of the report

REQUIREMENTS

- Class start at **1:15 pm**. Please be present in the classroom 5-10 minutes early to prepare, start up computers, read practice materials, etc.
- The **ACCT5123.P24.CTTT.1** practical sessions will begin on **March 11, 2025**, from **period 6 to period 10**, alternating weeks, and will take place in **room B3.08**.
- Tasks should be completed individually and in **groups (3 - 4 students per group).** Each group is to choose one topic for their course project.
- Each group will appoint a leader and submit their assignments on Moodle. Assignments can be resubmitted right after the class ends. **Late submissions** or **submissions via email** will be considered as **not submitted.**"

REQUIREMENTS

- Groups may refer to other topics and projects, but must understand the referenced topics and projects thoroughly. Absolutely **NO copying or modifying** the referenced topics and projects as your own. If discovered, all members of the group will receive a zero for the project.
- To announce, support, and promptly answer any information regarding the practical part of the course, students are kindly requested to join the class's Zalo chat group.
- **Contact information channels when needed:**
  - Zalo group/Personal Zalo
  - Email: khalvd@uit.edu.vn

DURATION AND GRADING RATIO

Course duration: **30 periods (Consisting of 6 weeks, 5 periods per week)**

Grading ratio: **50% of the final course grade.**

Evaluation content:

- ✓ In-class and homework assignments. **(40%)**
- ✓ End-of-term project (Group work). **(60%)**

CONTENT OF PRACTICE LESSONS

Week 1: Introduction to ERP & SAP S/4HANA

- Overview of ERP, SAP S/4HANA, and the Global Bike Group case study.
- Installing SAP GUI on Windows/MacOS.
- Navigating the S/4HANA interface.
- Creating master data: Customers, products, warehouse structure, price lists.

Week 2: Odoo & Master data

- Introduction to Odoo, comparison with SAP.
- Creating product categories, customers, and suppliers.
- Structuring warehouses & bin locations.
- Entering initial inventory data & opening balances for bank accounts

CONTENT OF PRACTICE LESSONS

Week 3: Sales - SAP SD & Odoo

- Sales & Distribution (SD) process in ERP
- Creating Quotations, Sales Orders, Delivery Orders, and Invoices in SAP (SD Module)
- Creating Opportunities, Quotations, and Sales Orders in Odoo

Week 4: Purchase & Inventory management - SAP MM & Odoo

- Materials Management (MM) process in ERP
- Creating Purchase Orders & Goods Receipt in SAP (MM Module)
- Creating RFQs, Purchase Orders, Delivery Orders, and Stock Count Records in Odoo

CONTENT OF PRACTICE LESSONS

Week 5: Production planning (PP) & Financial accounting (FA) - SAP PP & FA, Odoo

- Production planning (PP) process in ERP
- Practicing production planning in SAP (PP Module)
- Financial accounting (FA) process in ERP
- Creating customer invoices, vendor bills, credit notes, and debit notes in Odoo
- Recording Accounts payable (AP) & Accounts receivable (AR) in SAP (FA Module)

CONTENT OF PRACTICE LESSONS

Week 6: Human Resource Management (HCM) & Finalizing the Project – SAP HCM & Odoo

- Overview of Human Capital Management (HCM) in ERP.
- Creating employee records, HR management, and payroll processing in SAP (HCM Module).
- Data consolidation, validation, and final project reporting.

REQUIREMENTS FOR PROJECTS

Requirements:

- Each group will register one topic to apply ERP in practice. The project will be implemented on Odoo system.
- The project report includes:
  - Report file (following the provided template).
  - Demo on Odoo system.
  - All related documents.
- Additionally, each week of practical sessions will include individual assignments requested by the teaching assistant.

REQUIREMENTS FOR PROJECTS

Requirements:

- **TOPIC SELECTION**
  - Real-life scenario
  - Trending topic
  - Integration of multiple subjects.
- **RECOMMEND**
  - Electronic Commerce, Fashion, Food and Beverage Service, Beauty, Interior Design, Grocery, Sports, Home Appliances,...

REQUIREMENTS FOR PROJECTS

- **PROJECT REQUIREMENTS**
  - Manufacturing part not mandatory.
- **MASTER DATA**

✓ More than 20 product categories
✓ At least 500 products
✓ At least 1000 customers
✓ At least 1000 suppliers
✓ At least 5 users

✓ At least 3 warehouses
✓ At least 30 warehouse locations
✓ Complete initial inventory data for products
✓ Complete initial balances for Cash and Bank Ledgers.
✓ At least 50 customers / suppliers with opening balances.

REQUIREMENTS FOR PROJECTS

- **OPERATIONS:**
  - Ensure sufficient operational data across modules: CRM, Sales, Purchase, Inventory, Accounting

CRM

- ✓ Configure a complete series of Stages, Activities
- ✓ At least 100 Opportunities (at least 10 Opportunities in each stage)

SALES

- ✓ Create at least 100 Quotations
- ✓ At least 50 Sales Orders

REQUIREMENTS FOR PROJECTS

- **OPERATIONS:**

INVENTORY

- ✓ Create at least 50 Delivery Orders
- ✓ At least 50 Receipts
- ✓ At least 30 Stock Adjustments

PURCHASE

- ✓ Create at least 100 RFQs
- ✓ At least 50 Purchase Orders

ACCOUNTING

- ✓ Create at least 50 Customer Invoices
- ✓ At least 50 Vendor Bills
- ✓ At least 10 Credit Notes
- ✓ At least 10 Debit Notes
- ✓ Control payment activities for cash and bank accounts

REQUIREMENTS FOR PROJECTS

Another option: Research

- **Applying artificial intelligence to propose improvements, upgrading business processes for enterprises.**
- **Content implementation:**
  - Study theoretical foundations, draw diagrams of business processes, propose AI applications to optimize any business process.
  - Must develop a demo product for the proposal."
- **Recommend:** Supply chain optimization, Enhanced customer service, Production process optimization, Financial data analysis, Recruitment process optimization, Data management and security.

REQUIREMENTS FOR PROJECTS

Topic 1: AI-Based Demand Forecasting in ERP

- **Description:**
  - Utilize Machine Learning (ML) to forecast product demand based on historical sales data, market trends, and seasonal factors.
  - Integrate forecasting models into the Sales & Distribution (SD) Module in SAP or Odoo.
- **Implementation steps:**
  - Collect data from Sales Orders, Inventory, and CRM.
  - Apply Time Series Forecasting algorithms (ARIMA, LSTM, Prophet) to predict demand.
  - Integrate the results into the ERP system for automatic inventory and purchasing updates.

REQUIREMENTS FOR PROJECTS

Topic 2: Automating Invoice Processing with AI & RPA

- **Description:**
  - Combine AI & RPA (Robotic Process Automation) to automate invoice processing in the Financial Accounting (FA) Module.
  - AI extracts data from PDF invoices, and RPA inputs the data into the ERP system.
- **Implementation steps:**
  - Use OCR (Optical Character Recognition) to extract text from invoices.
  - Apply Natural Language Processing (NLP) to identify key details (invoice number, date, amount, vendor).
  - Use RPA (UiPath, Automation Anywhere) to automatically enter data into SAP/Odoo.

REQUIREMENTS FOR PROJECTS

Topic 3: AI Chatbot for Sales Process in ERP

- **Description:**
  - Develop an AI Chatbot to assist sales representatives in creating Quotations and Sales Orders in the ERP system.
  - The chatbot analyzes customer requests, suggests pricing, and generates sales orders in SAP SD Module.
- **Implementation steps:**
  - Use GPT-based AI (ChatGPT, Llama) to develop the chatbot.u
  - Connect the chatbot to the CRM & Sales system to retrieve customer data.
  - Integrate the chatbot into SAP Fiori or Odoo UI to streamline order processing.

REQUIREMENTS FOR PROJECTS

Topic 4: AI for Warehouse Optimization in ERP

- **Description:**
  - AI optimizes warehouse storage by automating product placement, reducing picking time in the Materials Management (MM) Module.
- **Implementation steps:**
  - Use Clustering algorithms (K-Means, DBSCAN) to group products based on sales frequency.
  - Apply Genetic Algorithms to optimize warehouse routes.
  - Integrate AI into the Warehouse Management System (WMS) in SAP/Odoo.

REQUIREMENTS FOR PROJECTS

Topic 5: Predicting Customer Payment Behavior Using Machine Learning

- **Description:**
  - Use ML to predict customers' ability to make timely payments based on their purchasing and payment history in the Financial Accounting (FA) Module.
  - Helps businesses reduce credit risk.
- **Implementation steps:**
  - Collect data from Customer Invoices, Payment History, and Credit Scores.
  - Apply Classification algorithms (Logistic Regression, Random Forest, XGBoost) to predict late payments.
  - Integrate results into the ERP system for credit risk assessment.

REQUIREMENTS FOR PROJECTS

Topic 6: AI for HRM - Employee Attrition Prediction

- **Description:**
  - Use ML to predict employee turnover based on historical work performance, job satisfaction, and attendance in the Human Capital Management (HCM) Module.
- **Implementation steps:**
  - Collect data from HR Records, Performance Reviews, Employee Satisfaction Surveys.
  - Apply Classification algorithms (SVM, Random Forest, Neural Networks) to analyze attrition factors.
  - Provide HR with predictive insights to retain employees.

REQUIREMENTS FOR PROJECTS

Topic 7: Integrating Blockchain into ERP for Supply Chain Management

- **Description:**
  - Integrate Blockchain into ERP to ensure transparency and security in Supply Chain Management (SCM).
- **Implementation steps:**
  - Develop Smart Contracts on Ethereum or Hyperledger to automate order verification and payments.
  - Connect Blockchain with SAP MM & Odoo Purchase Module to track transactions.
  - Use Blockchain to prevent fraud and enhance data integrity.

REQUIREMENTS FOR PROJECTS

Topic 8: AI-Powered Dynamic Pricing Optimization

- **Description:**
  - Use AI to optimize product pricing in real time based on supply-demand, competitor pricing, and market trends.
- **Implementation steps:**
  - Collect pricing data from Sales Orders, Market Trends, and Competitor Prices.
  - Apply Reinforcement Learning (Deep Q-Learning, PPO) to optimize pricing strategies.
  - Integrate AI-driven pricing into the SAP SD Module & Odoo Pricing Rules.

REQUIREMENTS FOR PROJECTS

Requirements:

- During the course, groups have 4 submission opportunities to report on the project progress:
  - **First submission:** Submit topic information, including a clear description of the objectives, scope, expected results, list of members, and the role of each member in the group.
  - **Second submission:** Submit the project execution plan: Plan - WBS (googling), Gantt Chart

REQUIREMENTS FOR PROJECTS

Requirements:

- During the course, groups have 4 submission opportunities to report on the project progress:
  - **Third submission:** Mid-term project progress report. Present directly in the practice class or record a short video to present the current progress of the project and the next steps, based on the plan outlined in the second submission (with adjustments, if any, update the plan accordingly).
  - **Fourth submission:** Submit all documents, report files of the topic.

REPORT CONTENT

PART 1: COVER PAGE

- *Includes course information, project title, instructor's name, and a list of team members.*

PART 2: SUMMARY

- *No longer than 2 A4 pages.*
- *Provides an overview of the report and the project's findings.*

PART 3: TASK ASSIGNMENT TABLE

PART 4: TABLE OF CONTENTS

PART 5: REPORT CONTENT STRUCTURE

- *Presents the main content report according to the suggested 4 chapters*

PART 6: REFERENCES

GENERAL STRUCTURE OF THE REPORT

CHAPTER 1: GENERAL INTRODUCTION

- *Overview of the ERP project, Scope, project objectives, Key stakeholders and personnel, etc.*

CHAPTER 2: THEORETICAL OVERVIEW

- *Theoretical basis of ERP, implementation platform, and problem description.*

CHAPTER 3: BUSINESS PROCESS SYSTEM

- *Introduction to the system overview.*
- *Introduction to the process including flowcharts, specifications, and illustrations.*

CHAPTER 4: CONCLUSIONS AND FUTURE DIRECTIONS

- *Advantages, limitations, and directions for development.*

QUESTIONS & ANSWERS

Contact help

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